Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 30 epochs with a batch size of 128.

2 Results

Figure 1 shows the test accuracy of DistilBERT by dataset.

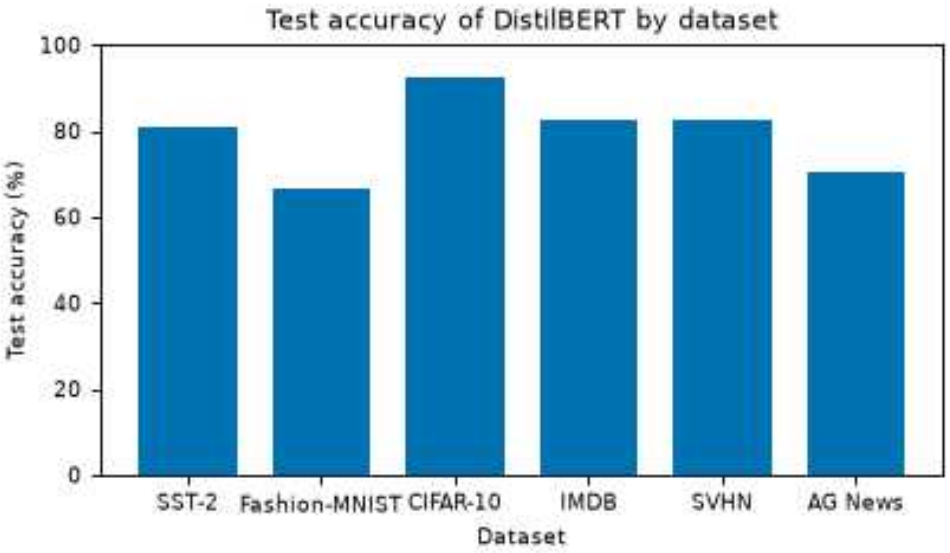


Figure 1: Test accuracy of DistilBERT by dataset for the models in this experiment.

DistilBERT achieves 66% test accuracy on Fashion-MNIST. DistilBERT scores 54.0 points higher on AG News than on Fashion-MNIST. DistilBERT scores 10.6 points higher on SST-2 than on AG News (Figure 1). DistilBERT scores 59.7 points higher on CIFAR-10 than on SVHN (Figure 1).

Figure 2 shows the accuracy by dataset.

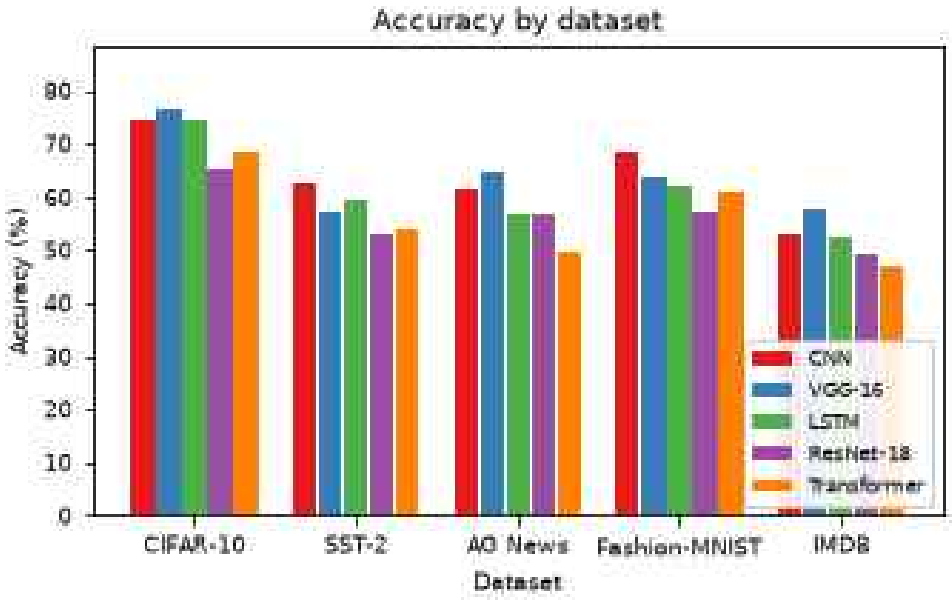


Figure 2: Accuracy by dataset for the models in this experiment.

As shown in Figure 2, on CIFAR-10, CNN is 6 points better than Transformer. ResNet-18 reaches 57.1% accuracy on SST-2. On AG News, ResNet-18 is about 7 points better than Transformer. ResNet-18 reaches 56.7% accuracy on AG News.

Figure 3 shows the training loss.

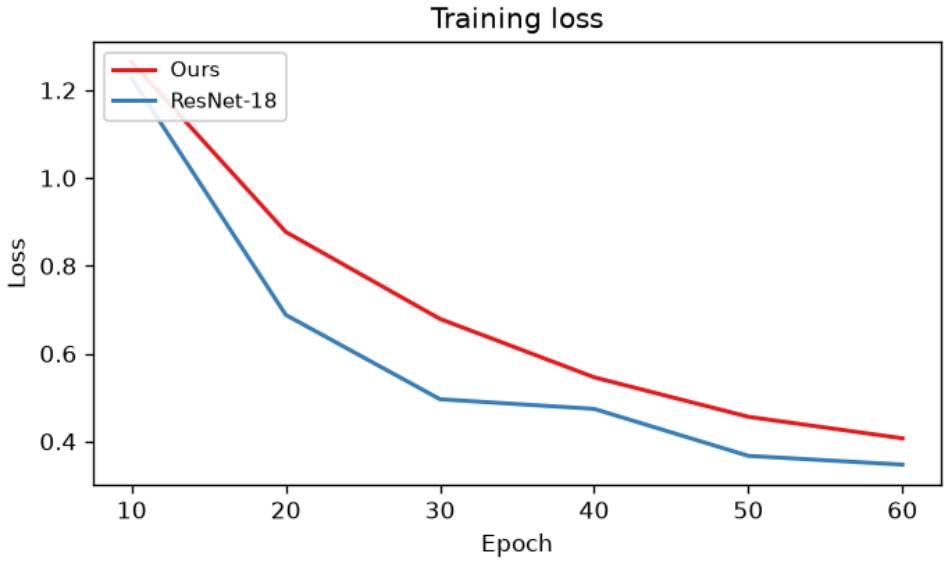


Figure 3: Training loss for the models in this experiment.

ResNet-18 has a loss of about 0.5 at epoch 30. As shown in Figure 3, Ours has a loss of 0.729 at epoch 30. At epoch 50, the loss of ResNet-18 is 0.456.

Figure 4 shows the training loss.

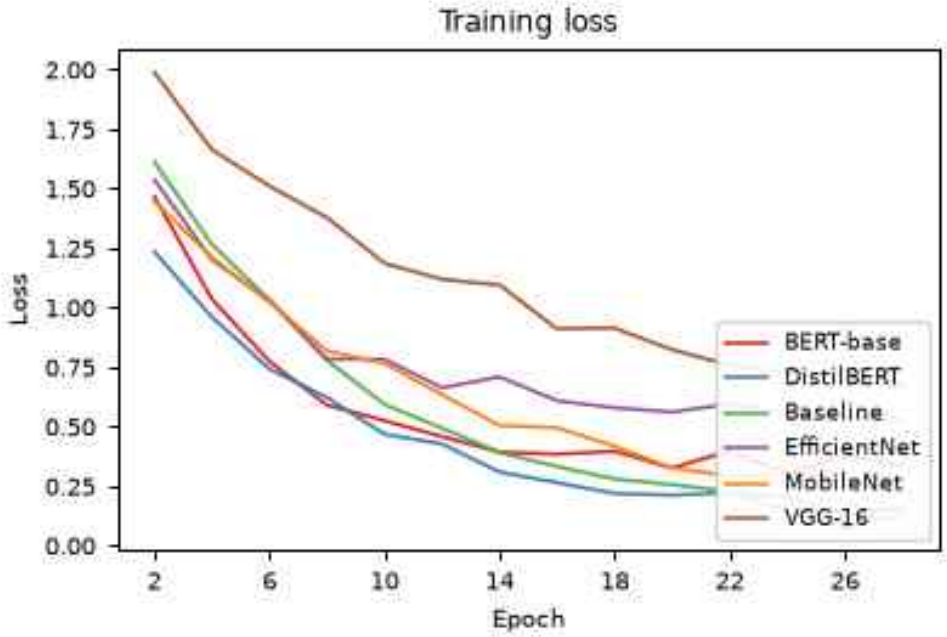


Figure 4: Training loss for the models in this experiment.

At epoch 10, the loss of VGG-16 is 1.186. At epoch 2, the loss of Baseline is 1.61.

3 Discussion

We note that the training was stable for all models.